

New biomarkers for the diagnosis of primary central nervous system lymphoma in CSF: A multicenter retrospective cohort study

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Abstract

Background. Primary central nervous system lymphoma (PCNSL) is a rare form of non-Hodgkin lymphoma. Diagnosis usually requires a brain biopsy, which is associated with complications and delay of diagnosis. Cerebrospinal fluid (CSF) analysis can provide a diagnosis without the need for a biopsy, but only in a minority of PCNSL patients. The aim of this study is to identify new potential biomarkers in CSF for patients with clinical suspicion of PCNSL to improve the diagnostic yield of CSF analysis.

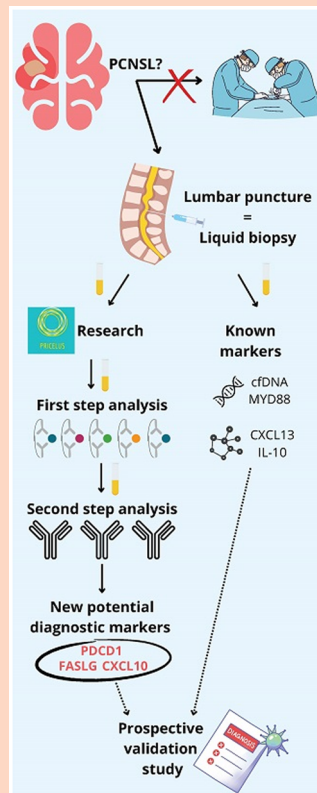
Methods. CSF samples were analyzed from a retrospective cohort that included patients with clinical suspicion of PCNSL. First-step analysis consisted of targeted multiplex biomarker analysis with use of a proximity extension assay (Olink), allowing the simultaneous analysis of relative expression levels of 183 protein markers in a small CSF volume (1 µl). To validate the potential identified biomarkers, a second quantitative analysis was performed in a subgroup with a bead-based multiplex immuno-assay.

Results. We included 74 patients, of whom 11 patients (15%) were diagnosed with PCNSL. In the primary analysis, 30 markers showed significant differences ($P < .05$) between PCNSL patients and controls. After correction for multiple testing, 6 markers remained significant. The quantitative analysis confirmed significant differences ($P < .05$) for 3 markers: CXCL10, programmed cell death protein 1 (PDCD1), and FAS natural ligand.

Conclusions. We identified a panel of novel potential diagnostic markers in CSF for the diagnosis of PCNSL and confirmed the value of several known markers. The diagnostic value of these markers will be validated in a multicenter prospective study.

Key Points

- Diagnostic value of modern CSF markers in suspected PCNSL demonstrated in multicenter cohort
- Several chemokines and cytokines require further prospective validation: CXCL10, PDCD1, and FASLG.

Graphical Abstract**Importance of the Study**

Primary central nervous system lymphoma (PCNSL) continues to be diagnosed predominantly via brain biopsy, an invasive and risk-carrying procedure that often leads to delayed diagnosis. To enhance the diagnostic utility of cerebrospinal fluid (CSF), novel liquid biopsy approaches offer promise, particularly those using cell-free DNA and chemokine profiling. In a retrospective cohort study of 74 consecutive patients with suspected PCNSL, we confirmed the diagnostic value of known markers and identified several new potential diagnostic biomarkers.

Through a two-step analysis of marker finding and initial validation, we constructed a promising CSF biomarker panel that incorporates known markers and introduces 3 novel candidates: CXCL10, programmed cell death protein 1 (PDCD1), and FAS ligand (FASLG). These results provide a foundation for prospective multicenter validation aimed at advancing noninvasive diagnostic strategies for PCNSL. If validated, this may improve outcomes for PCNSL patients through faster, less-invasive diagnosis and more rapid initiation of therapy.

Primary central nervous system lymphoma (PCNSL) is a relatively uncommon, aggressive type of extranodal non-Hodgkin lymphoma. Unlike many other high-grade brain tumors, PCNSL has the potential to be cured, with long-term survival or cure in 40% of patients following intensive treatment.^{1,2} While surgical removal of tumor tissue is typically a crucial step in the diagnosis and initial treatment of most brain tumors, surgery plays a limited role in the treatment of PCNSL.³ When there is suspicion of PCNSL, a stereotactic brain biopsy is often performed. Brain biopsy carries inherent risks, including a risk of serious

complications (eg, intracranial hemorrhage) of ~4%-8% and a mortality rate between 0.8% and 2.2%.⁴⁻⁶ In a minority of patients, alternative methods such as the examination of vitreous fluid (identification of lymphoma cells) or cerebrospinal fluid (CSF) can be utilized to diagnose PCNSL, thereby obviating the need for brain biopsy.

The current role of CSF diagnostics in PCNSL is limited to cytology and flow cytometry, which leads to a diagnosis in only a small proportion of patients (less than 25%).⁷ Recent research indicates a potential role for new CSF markers in the diagnosis of PCNSL, including myeloid differentiation

primary response (88) (MYD88) L265P mutation in cell-free (cf)DNA, levels of C-X-C Motif Chemokine Ligand 13 (CXCL13), and interleukin 10 (IL10).⁸ Although recent guidelines consider these new techniques promising,⁹ none of the available CSF markers have been sufficiently studied to support clinical decision-making.¹⁰

We performed an exploratory multicenter retrospective cohort study to identify new biomarkers in CSF for the diagnosis of PCNSL and performed validation of their clinical diagnostic value. Through the use of efficient assays that allow for simultaneous testing of many potential markers, we aimed to maximize the diagnostic potential of CSF analysis. As such, this line of research seeks to improve the diagnostic process of PCNSL and ultimately reduce the necessity of a stereotactic biopsy to obtain a tissue diagnosis.

Methods

Study Cohort

The retrospective study cohort consists of consecutive patients who were suspected of PCNSL at 2 Dutch hospitals, University Medical Center Utrecht (an academic hospital) and St Antonius Hospital (a large non-academic hospital), between August 2020 and October 2021, for whom CSF samples were available. Patients were identified by electronic records of CSF analysis, including flow cytometry. The inclusion criteria for this study were: (1) patients, aged 18 years and older, with clinical suspicion of PCNSL, as determined by the treating physician based on clinical presentation and imaging; and (2) CSF samples, collected and stored as part of routine patient care prior to treatment, were accessible for analysis. Patients with lymphoma manifestation outside the central nervous system at the time of CSF sampling were excluded from the study. After informed consent, materials were collected and analyzed.

Baseline Characteristics

We collected the following baseline characteristics from the electronic medical record: age at diagnosis, final diagnosis, sex, results of routine CSF analysis (leucocytes, erythrocytes, total protein, glucose), results of cytopathological analysis of CSF (detection of neoplastic cells), results of CSF flow cytometry, and results of potential new biomarkers tested in CSF, in case these were collected as part of routine clinical care in individual cases (IL10, CXCL13, and MYD88 analysis on cfDNA). IL10 and CXCL13 were measured with use of a bead-based multiplex immunoassay based on Luminex technology.¹¹ The MYD88 mutation detection in CSF was determined by means of a droplet digital polymerase chain reaction (PCR) mutation analysis, as reported earlier.¹²

As a reference test, a final diagnosis of PCNSL was made by a positive brain biopsy of the lesion (evaluated by a neuropathologist), CSF analysis (monoclonal expansion detected by immunophenotyping and/or identification of malignant cells through cytological examination in patients with typical MRI findings of PCNSL), or diagnosis of vitreoretinal lymphoma through vitreous fluid analysis (in patients with typical MRI findings of PCNSL), in line with contemporary guidelines.¹³

Analysis

As part of clinical care, CSF samples (1 to 5 ml) were collected by lumbar puncture (or by an Ommaya reservoir in 1 case), directly processed (centrifuging at 2000 g for 10 min), and stored at -80°C . Samples underwent a single freeze-thaw cycle prior to initial analysis, during which a minimal volume (1 μl) was pipetted before samples were immediately refrozen. For the second analysis, previously available material was thawed once more and analyzed directly.

The initial analysis involved targeted multiplex biomarker analysis with the proximity extension assay (PEA) by Olink Bioscience, Uppsala, Sweden.¹⁴ With PEA, antibody pairs targeting the same protein of interest are coupled with matching DNA sequences. After recognition of the protein of interest, thus being in close proximity of each other, these DNA sequences will hybridize. With the use of real-time PCR amplification, the relative expression levels of multiple proteins in a small volume of CSF (1.2 μl) are obtained. We used 2 panels for the analysis: (1) the immuno-oncology panel, consisting of 92 proteins related to immunological processes and immuno-oncological conditions; and (2) the neuro-exploratory panel, also consisting of 92 proteins related to neurological conditions or processes.¹⁵ There was 1 protein present in both panels (IL15).

The outcomes of the primary (Olink) analysis are given in an arbitrary unit used by Olink, called Normalized Protein eXpression (NPX), which are on a logarithmic scale (Log2). One NPX difference means a doubling of the protein amount within the samples. It is calculated based on actual values and data preprocessing (normalization) is conducted to minimize both intra- and inter-assay variation. No data were excluded, even if values were below the limit of detection (LOD). The LOD is based on 3 times the deviation of the results from negative controls. Using data below the LOD is acknowledged as a reliable strategy for increasing power, as it does not increase the number of false-positive results.¹⁶

To validate the diagnostic potential of the biomarkers from this PEA analysis in a quantitative manner, a second analysis was performed in a subgroup of patients with a bead-based multiplex immuno-assay based on Luminex technology.¹¹ This analysis was done in a subgroup of patients who were selected from the initial study group and consisted of patients presenting with (space occupying) lesions with a high degree of suspicion of having PCNSL. For this second analysis, only a limited number of markers could be investigated due to the limited amount of available CSF. We chose potential markers based on: (1) whether values were significantly different between PCNSL and non-PCNSL cases in the Olink analysis (CD27, CXCL13, FAS natural ligand [FASLG], CXCL10, programmed cell death protein 1 [PDCD1], tumor necrosis factor receptor superfamily member 9 [TNFRSF9]); (2) whether they were potentially new markers, that is, not extensively previously reported as a potential marker for PCNSL (thereby excluding CXCL13 for the second analysis); (3) whether the marker could be analyzed with available technology (thereby excluding TNFRSF9 for the second analysis); and (4) whether potential markers were independent or related, based on the protein-protein interaction network using the STRING database (including interferon (IFN)-gamma into the second analysis).¹⁷

Statistical Analysis

Differences in baseline characteristics for PCNSL and non-PCNSL patients were evaluated with a chi-square test.

After checking the results of the primary (Olink) analysis for normality (based on histograms), differences in NPX values between PCNSL patients and controls were determined with the use of a two-sample *t*-test. These first results were adjusted for multiple testing with the Benjamini–Hochberg method. The significance level was set at $P < .05$. Data were visualized with a volcano plot. Additionally, a principal component analysis (PCA) was performed for unsupervised analysis of the data.

The dataset obtained with the multiplex immuno-assay consisted of nonparametric continuous data, which were analyzed with Mann–Whitney *U*-tests.

We performed statistical analyses in R (version 4.2.2.), SPSS (version 26.0, IBM SPSS Statistics), and Olink Insights Stat Analysis (Olink Statistical Analysis app).

Standard Protocol Approvals, Registrations, and Patient Consents

Ethical approval was granted by the UMC Utrecht Biobank Research Ethics Committee (protocol 14-646). Participants provided written informed consent to participate in the study. Of note, acquisition of CSF was done as part of routine clinical care and not primarily for the purpose of this study.

Data Availability

Anonymized data, not published within this article, will be made available by reasonable request from any qualified investigator.

Results

Demographics

The study population comprised 74 patients, of whom 11 (15%) were ultimately diagnosed with PCNSL. The final

PCNSL diagnosis was histopathologically confirmed in 8 of 11 patients. Two patients were diagnosed via CSF analysis (both exhibited monoclonal expansion detected by immunophenotyping and malignant cells identified through cytology), and 1 through diagnosis of a vitreoretinal lymphoma. Corticosteroids were initiated in 6 out of 11 (55%) patients ultimately diagnosed with PCNSL before lumbar puncture and in 7 out of 63 (11%) patients ultimately diagnosed without PCNSL. No patients were HIV-positive or otherwise immunocompromised. Baseline characteristics are presented in [Table 1](#). The non-PCNSL patient group included patients diagnosed with metastases (lung and sinonasal carcinoma), gliomas (glioblastoma, IDH-wildtype), inflammatory diseases (mainly multiple sclerosis), and infectious diseases (meningitis, brain abscess). Furthermore, a wide range of other diagnoses was observed, such as cerebral ischemia, neuromuscular disorders, and patients with intracranial lesions or polyradiculopathy without a definitive diagnosis. There was no significant difference between patients and controls in terms of age and sex. Although potential new biomarkers were analyzed (as part of clinical routine) in just a minority of patients, CXCL13, IL10, and MYD88 mutation analysis correlated with the clinical diagnosis of PCNSL. There was 1 patient without PCNSL in whom a MYD88 mutation was detected, but when this analysis was repeated after 1 month, it yielded a negative result for MYD88. A final diagnosis of multiple sclerosis was made during follow-up. After 4 years of follow-up, no evidence of malignant lymphoid disorder has been observed.

Identification of Potential New Biomarkers for PCNSL Diagnosis

A total of 183 proteins were assessed for 74 patients. From the 183 measured proteins, 30 proteins differed significantly between patients diagnosed with PCNSL and controls ([Table 2](#) [significant results only], [Figure 1](#) Volcano plot and [Table S1](#) [full results]). There was a positive correlation between the NPX values of the marker and the probability of diagnosing PCNSL for most markers (28/30 proteins). This set of potential markers contains 4 known potential markers (CD27, CXCL13, CXCL9, and IL10) as well as 26 markers that had

Table 1. Baseline characteristics

	Non-PCNSL	PCNSL	<i>P</i> -value	Non-PCNSL: metastases/gliomas	Inflammatory/ infectious diseases	Other
<i>n</i>	63	11		4	12	47
Male (%)	38 (60)	8 (73)	.43	3 (75)	10 (83)	25 (53)
Age, y (IQR)	64 (52-70)	68 (56-75)	.39	63 (49-74)	67 (53-69)	63 (51-70)
Leukocytes ^a	11/63	7/10	.01	2/4	6/12	3/47
Protein ^a	48/62	10/11	.31	4/4	11/12	33/46
CXCL13 ^a	0/6	4/5	.02	0/1	-	0/5
IL10 ^a	0/7	5/5	.01	0/1	0/1	0/5
MYD88	1/20	4/8	.01	0/2	0/4	1/14

Abbreviation: IQR, interquartile range.

^aElevated leukocytes > $5 \times 10^9/L$; protein > 0.40 g/L; CXCL13 > 90 pg/mL¹⁸; IL10 > 16 pg/mL¹⁸.

Table 2. Obtained results for the exploratory analysis showing only the significantly different proteins between PCNSL and non-PCNSL

Marker	Panel	non-PCNSL (NPX, mean)	PCNSL (NPX, mean)	P-value, uncorrected	Adjusted P-value
CD27 ^a	IO	4.70	8.35	.000004	.0007
CXCL13	IO	4.53	9.07	.00001	.0007
FASLG ^a	IO	4.31	6.04	.00001	.0007
CXCL10 ^a	IO	7.13	9.87	.0002	.009
PDCD1 ^a	IO	2.57	5.02	.0002	.009
TNFRSF9	IO	3.13	7.10	.0006	.02
CD5	IO	1.93	3.17	.003	.09
ADA	IO	4.12	6.21	.003	.09
CXCL11	IO	1.87	3.19	.005	.10
CXCL9	IO	4.24	6.27	.005	.10
CD8A	IO	3.11	4.35	.006	.10
CCL3	IO	4.15	5.12	.008	.10
IL10	IO	3.48	6.20	.01	.11
MCP-2	IO	4.29	5.34	.01	.13
NEFL	NE	8.40	10.41	.01	.13
GPMB	NE	4.95	5.65	.01	.13
IFN-gamma ^a	IO	3.93	5.62	.01	.13
HO-1	IO	4.13	6.28	.02	.17
CD40	IO	7.63	8.73	.02	.17
CCL19	IO	6.95	8.30	.02	.17
Gal-9	IO	6.30	6.97	.02	.17
IL-1 alpha	IO	0.45	0.06	.02	.17
CRTAM	IO	2.60	3.35	.02	.17
CCL23	IO	2.88	3.85	.02	.17
CCL17	IO	2.51	4.01	.03	.20
IFI30	NE	3.03	3.70	.03	.22
IL12	IO	3.26	4.17	.03	.22
GZMA	IO	4.55	5.34	.04	.24
MUC-16	IO	1.68	1.29	.04	.26
MIC-A/B	IO	4.01	4.66	.04	.26

Abbreviations: IO, immuno-oncology panel; NE, neuro-exploratory panel; NPX, normalized protein eXpression.

^aSelected for validation in second analysis.

not been described previously for the diagnosis of PCNSL. After adjusting for multiple testing, 6 potential markers remained significant: CD27, CXCL13, FASLG, CXCL10, PDCD1, and TNFRSF9.

To investigate whether we could identify patient clusters based on the proteomics data, we used PCA (Figure S1). Principal component (PC) 1 and PC 2 explained more than 60% of the variance between the samples. K means clustering did not reveal clusters containing only samples of 1 of the 4 patient groups (PCNSL; metastases/gliomas; inflammatory/infectious diseases; other). We determined 5 clusters as the optimal number of clusters using the elbow method (Figure S2). PCA plot with 5 k means clusters showed no potential group of patients (Figure S3).

Second-Step Analysis: Multiplex Immunoassay

For validation of the potential biomarkers found in the exploratory analysis, a validation of these markers was performed by measuring these proteins using a bead-based multiplex immunoassay in a subgroup of 32 patients, a selected subset of the original study cohort with high clinical suspicion of PCNSL. This validation cohort consisted of 11 patients with PCNSL and 21 controls. See Table S2 for their final diagnosis.

Five potential biomarkers were chosen for this validation analysis: CD27, FASLG, CXCL10, PDCD1, and IFN-gamma. For 3 of these potential biomarkers, significant differences between PCNSL patients and controls ($P < .05$) were found in this second, confirmatory analysis: CXCL10, PDCD1, and FASLG (Figure 2 and Table S2). Levels of IFN-gamma were not detectable in CSF with the multiplex immunoassay (below LOD). For CD27, no significant difference was found ($P = 1.00$), which is probably explained by missing levels (below LOD) in 23 out of the 32 samples (Figure 2 and Table S2). Table S3 shows that stratifying by corticosteroid treatment reveals similar trends in patients with versus patients without corticosteroid use. However, most associations no longer reach statistical significance, likely due to small sample sizes per subgroup.

Discussion

In this multicenter retrospective study, we identified several new potential markers for the diagnosis of PCNSL in CSF. We also confirmed the diagnostic value of markers that were previously reported on, such as CXCL13.¹⁸ In the first exploratory (Olink) analysis, 30 biomarkers showed a significant result, with 6 showing a significant result after adjusting for multiple testing. Afterwards, we confirmed the diagnostic potential for 3 potential markers in a quantitative manner (multiplex immunoassay analysis): FASLG, CXCL10, and PDCD1. Although our primary aim was to discover new (protein-based) markers for the use of PCNSL diagnosis, we also analyzed the clinically acquired data on previously described markers, including CXCL13, IL10, and cfDNA analysis of MYD88. These data, although underpowered, underline the potential use of these markers as a diagnostic marker. We will include these 3 previously identified biomarkers, as well as the recent discovered biomarkers from this report, in a prospective diagnostic study.

Looking into the new significant biomarkers that show high potential, we see that most of them (CD27, FASLG, and TNFRSF9) belong to the tumor necrosis factor (TNF) family, having a role in both T-cell and B-cell activation.¹⁹⁻²² CD27 has been described as a potential biomarker for PCNSL in 2000²³; however, the diagnostic value has never been confirmed in other cohorts. FASLG and TNFRSF9 have not been previously described as a biomarker for PCNSL. CXCL10 is a chemokine involved in the recruitment and activation of the immune system for tumor suppression²⁴ and has been described as a biomarker for several oncological and neurological conditions, but not yet as a biomarker for PCNSL. PDCD1 is a protein that can be found on T-cells and plays an

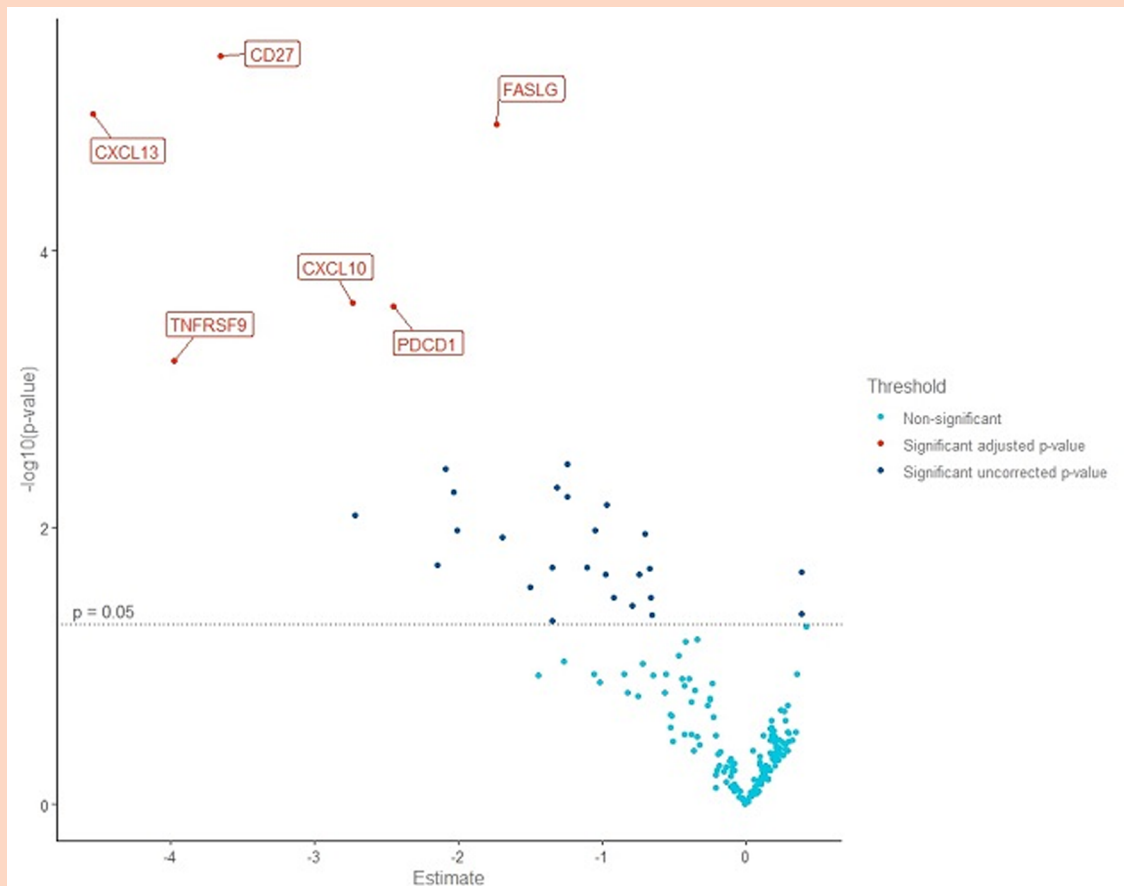


Figure 1. Volcano plot illustrating the results of the two-sample *t*-test; the x-axis displays the estimated difference; the y-axis displays the negative $\log_{10}$ *P*-value; the horizontal dotted line displays a *P*-value of .05.

important role in T cell exhaustion.²⁵ It has been extensively described in the context of PCNSL: as a prognostic biomarker in serum,²⁶ as a prognostic biomarker by immunostaining of tumor cells²⁷ and as a potential target of the tumor microenvironment acting as a potential targeted therapy.^{28,29} Until now, though, PDCD1 had not yet been described as a diagnostic marker in CSF.

For our study, 183 potential biomarkers were simultaneously analyzed in CSF of a diagnostic cohort of patients with a clinical and radiological suspicion of PCNSL. This is unlike other larger studies,^{8,18} in which only a small number of potential biomarkers were determined in selected cohorts of patients diagnosed with PCNSL. These previous studies typically used separate index and control cohorts, wherein the control datasets were not selected based on a clinical suspicion of PCNSL. This can result in exaggeration of the diagnostic value by contrasting PCNSL patients with non-relevant controls. By including a consecutive series of patients (with accessible CSF samples) from the appropriate clinical domain—patients with a clinical suspicion of PCNSL—we aimed for maximal validity of our current findings for clinical practice. The composition of the control group (“PCNSL mimics”) was determined by the clinical population in the 2 participating centers (1 academic and 1 non-academic)

and may differ from the patient population in other PCNSL centers; although such differences in populations may influence the diagnostic value of diagnostic markers, we aimed to optimize robustness of our results through a two-step analysis. Another strength of our study is the validation analysis with use of another technique (multiplex immunoassay), confirming FASLG, CXCL10, and PDCD1 as potential markers for clinical application. Based on the first exploratory (Olink) analysis, CD27 showed the highest potential. This could not be confirmed in the multiplex immunoassay due to values below the detection limit (of about 4 picograms per milliliter) and the limited amount of available CSF. A distinct (ideally prospective cohort) study is required for further clinical validation and to determine the markers’ individual and combined diagnostic test accuracy.

This retrospective cohort study has several limitations. The sample size of patients diagnosed with PCNSL in this study was relatively small, with only 15% of patients ultimately diagnosed with PCNSL. The proportion of patients with a final PCNSL diagnosis among those suspected in the literature ranges from 14% to 58%.^{30,31} Whereas these studies often provide limited details about their inclusion criteria, ours are clearly defined: patients undergoing lumbar puncture with flow cytometry for possible PCNSL. The

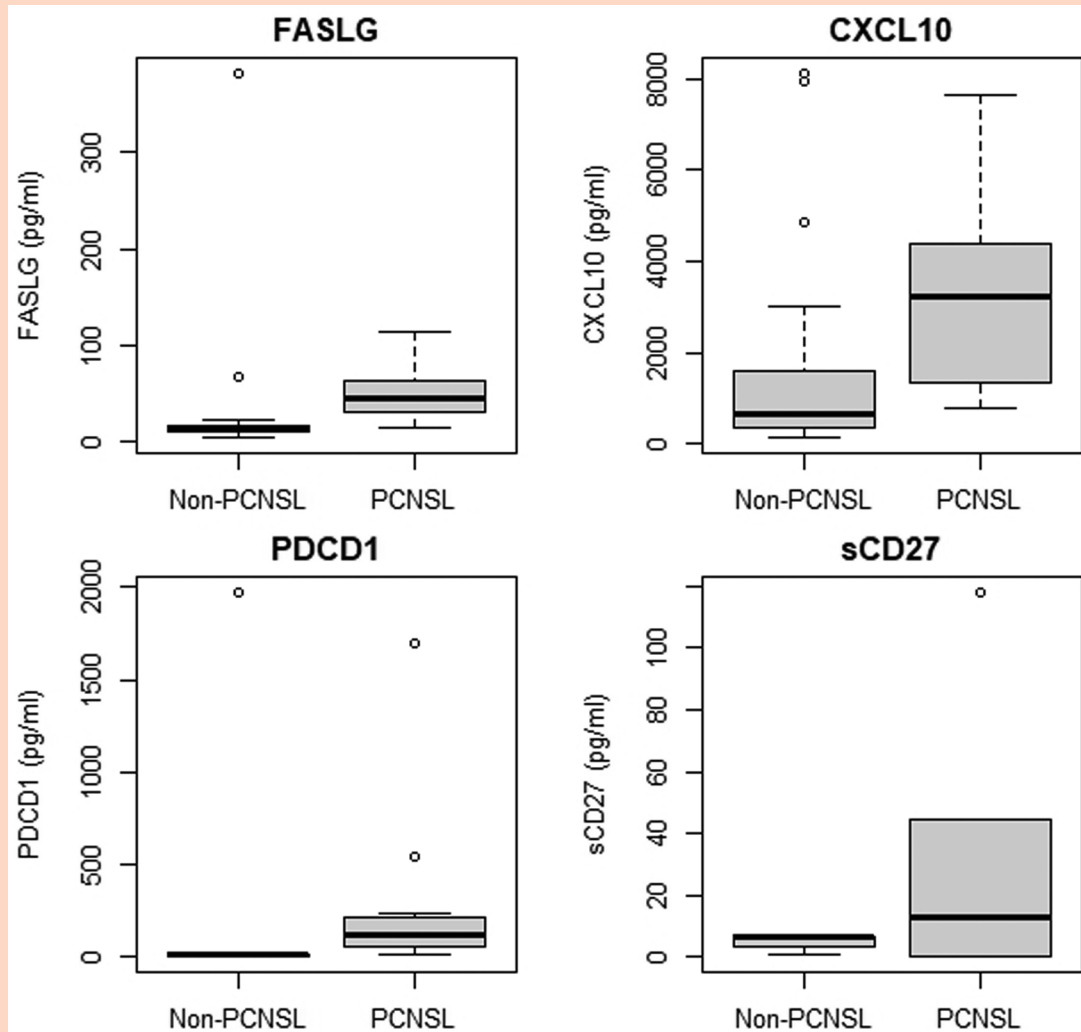


Figure 2. Multiplex immunoassay analysis of FASLG, CXCL10, PDCD1, and CD27.

control group included a number of patients from our academic neuromuscular clinic with low to moderate clinical suspicion of neurolymphomatosis, who underwent CSF analysis, including flow cytometry. For future clinical validation studies, separate analyses could be considered according to the degree of suspicion. Our current study was not designed or powered for this purpose. An additional limitation is that, due to the small sample size, we were unable to conduct a stratified analysis to assess diagnostic value in subgroups or the diagnostic impact of corticosteroids. Due to missing data and small sample sizes, we were also unable to accurately establish the diagnostic value for the novel biomarkers in a multivariable analysis. Consequently, we could not examine the added diagnostic value of the new biomarkers to other markers of interest, such as cfDNA analysis (MYD88).¹² The diagnostic cohort is not a fully consecutive group of patients suspected of clinical and radiological suspicion of PCNSL, since some patients could not be included as no CSF samples were accessible for analysis (due to minimal availability) or a lumbar puncture

was omitted in a clinical setting (eg, in the case of space-occupying lesions with the risk of cerebral herniation). As such, this cohort represents a selected subgroup of potential PCNSL patients for whom CSF could be obtained. However, studying this subgroup does reflect clinical practice, as the studied CSF markers will only be tested in these patients; as such, our results form an unbiased reflection of the diagnostic values of CSF markers in the subgroup with CSF availability.

Future research into liquid biopsy for the diagnosis of PCNSL is promising, but also leads to several new challenges. With the rapid developments in molecular diagnostics in hemato-oncology, it is likely that new molecular-genetic aberrations will be found with diagnostic, prognostic, or predictive value in PCNSL. Consequently, new studies for associated liquid biopsy markers in CSF or blood will have to be performed, and diagnostic panels will need to be updated accordingly. By establishing the diagnostic value of currently available markers, we aim to provide a baseline set to which new markers may be added.

The sensitivity of molecular-genetic analysis, currently limited to MYD88 L265P mutation detection in cfDNA, ranges from 52% to 92%, with larger cohorts generally showing lower sensitivity. Our findings are consistent, as we detected the mutation in 4 out of 8 PCNSL patients. The chemokines and cytokines analyzed in routine clinical practice showed improved sensitivity, with elevated IL10 and CXCL13 levels in 80% to 100% of patients. Ferreri et al.⁸ also demonstrated the added value of chemokine and cytokine testing, increasing sensitivity from 72% with MYD88 analysis alone to 94% when combined with IL-10 measurement. The exact diagnostic value of the new chemokines and cytokines could not be determined in the initial analysis, as the outcome was expressed in arbitrary units (NPX). In the second analysis, there was a small sample size, and a considerable amount of data was missing, making it impossible to reliably assess diagnostic accuracy without introducing multiple sources of bias and risking overfitting. Given the exploratory nature of this study, we plan to address these questions in more depth in an ongoing prospective diagnostic study.

Also, any biomarker finding study carries the risk of false-positive results and/or findings that only pertain to the specific patient group that was selected in the study. To overcome these risks of bias, a validation study is ideally powered adequately, with clinically relevant and reproducible inclusion criteria, validated diagnostic tests, and a multicenter setting to account for differences in local case mix and diagnostic process. That is why we are performing a prospective diagnostic study, in which we want to validate the diagnostic value of new biomarkers from this study, as well as previously identified biomarkers such as cfDNA analysis for the MYD88-mutation, both as separate markers and as a combined diagnostic panel. This combination of techniques could serve as an optimal diagnostic panel to enhance the diagnostic yield of CSF analysis for PCNSL. In this larger-scale study, the influence of corticosteroid use and the diagnostic value for subgroups (eg, with an well-defined intracranial mass) can be evaluated.

Conclusion

In this retrospective cohort study, we identified new potential diagnostic markers in CSF for the diagnosis of PCNSL and confirmed the value of previously described markers. The findings from this study open avenues for further research and, if validated, could be integrated into routine clinical practice. Such an innovation will increase the percentage of patients diagnosed with PCNSL based on CSF analysis without the need of biopsy, thereby facilitating timely treatment, avoiding biopsy-related complications, and potentially improving the outcome of PCNSL patients.

Supplementary Material

Supplementary material is available online at *Neuro-Oncology Advances* (<https://academic.oup.com/noa>).

Keywords

CSF biomarkers | CXCL10 | FASLG | PDCD1 | primary CNS lymphoma

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Conflict of Interest Statement

None declared.

Author Contributions

Conception and Design: J.L.M.v.R., P.H.W., T.J.S., M.C.M., S.N., and T.J.S. Data Collection and Preparation: J.L.M.v.R., P.H.W., E.M.D., S.N., and T.J.S. Data Analysis and Interpretation: J.L.M.v.R., P.H.W., E.M.D., B.M., H.J.T.R., T.S., M.C.M., S.N., and T.J.S. Writing—Original draft: J.L.M.v.R., P.H.W., and T.J.S. Writing—Review & Editing: J.L.M.v.R., P.H.W., E.M.D., B.M., H.J.T.R., T.J.S., M.C.M., S.N., and T.J.S.

Data Availability

Data are available from the corresponding author upon reasonable request.

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